

PRODUCT REQUIREMENT DOCUMENT (PRD)

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1. Project Overview

Problem Statement	Policy analysts and civil servants rely on slow, fragmented, manual processes to assess the impact of new Malaysian policies — requiring days of coordination across departments and expensive consultants.
Target Domain	Agentic AI workflow automation applied to Malaysian public policy analysis and wargame simulation.
Proposed Solution	AUTO-WARGAME lets any user describe a Malaysian policy in plain English and instantly receive a structured, six-stage wargame simulation — covering stakeholder impacts, economic effects, risk scoring, scenario comparison, and final recommendations — all orchestrated by GLM.

2. Background & Business Objective

Background

Traditional policy analysis in Malaysia follows a slow linear pipeline: drafting, specialist review, stakeholder consultation, and manual synthesis — often taking weeks. This process is inaccessible to smaller agencies and researchers without specialist resources.

Why This Matters

- Compresses policy impact analysis from days to minutes, enabling rapid iteration before decisions are committed.
- Makes structured policy wargaming accessible to non-specialist analysts, students, and researchers.
- GLM's agentic orchestration applies a consistent analytical framework to every simulation, reducing human bias.

Strategic Fit

AUTO-WARGAME directly addresses the UMHackathon 2026 Domain 1 challenge. Removing the GLM component collapses the workflow entirely — the six simulation stages cannot be coordinated by any other component.

3. Product Purpose

3.1 Main Goal

To provide an AI-native policy wargame engine — any user inputs a Malaysian policy scenario and receives a structured multi-stage simulation report within ~60 seconds.

3.2 Target Users

- Policy Analysts & Civil Servants — rapid, structured impact assessments without manual cross-department coordination.
- Researchers & Students — interactive exploration of real Malaysian policy scenarios.
- Journalists & Public Affairs Professionals — quick, data-backed understanding of proposed legislation.

4. System Functionalities

4.1 How It Works

The user submits a policy scenario (free text or preset) via the frontend. The Express.js backend orchestrates six sequential GLM calls — each stage receives prior stage output as context — and streams results live to the frontend via Server-Sent Events (SSE).

Stage	Name	What GLM Does
1	Policy Intake	Parses input; extracts policy instrument, target population, agency, and scope. Flags ambiguous inputs.
2	Stakeholder Analysis	Identifies affected groups (B40, M40, T20, SMEs, transport sector) and generates per-group impact assessments.
3	Economic Modelling	Estimates inflationary pressure, fiscal impact, GDP effects, and consumption changes.
4	Risk Scoring	Produces a risk matrix: social unrest, political feasibility, implementation complexity, fiscal risk.
5	Scenario Comparison	Generates two policy variants and compares them against the original across key metrics.

6	Report Generation	Synthesises all prior outputs into: Executive Summary, Key Risks, Recommendations, Alternative Scenarios.
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4.2 Key Features

- Six-stage GLM-orchestrated policy simulation with real-time SSE streaming.
- Five preset Malaysian policy scenarios for instant demonstration.
- Backend /health endpoint with API key validation shown as a green/red status pill.
- Graceful error handling — API key missing, model errors, oversized input, concurrent request queuing (HTTP 429).

4.3 AI Model & Prompt Design

Model: Z.AI ILMU-GLM-5.1, chosen for its strong instruction-following, sufficient context window for cumulative stage outputs, and reliable system-prompt role-locking.

Prompting Strategy: Multi-step agentic prompting — each stage has a dedicated system prompt and receives the original input plus all prior stage outputs as context. This sequential chain-of-thought approach is required because policy wargaming is inherently ordered (stakeholder impacts must precede economic modelling, risk scoring must precede scenario comparison).

Input Handling: Maximum 1,500 tokens per stage prompt. Inputs over this limit are automatically chunked across Stages 1–3 and synthesised before Stage 4. Inputs under 20 words are rejected at the frontend before any GLM call.

Fallback Behaviour: Malformed JSON triggers up to 2 retries with a stricter prompt. API failures retry up to 3 times. If retries fail, the stage is marked 'Output degraded — manual review recommended' and the workflow continues. Suspected hallucinated values (numbers absent from the original input) are flagged [UNVERIFIED] in the frontend.

5. User Stories & Use Cases

5.1 User Stories

User Role	I want to	So that...	Priority	Feature
Policy Analyst	submit free-text policy and get a structured impact report	I can produce a rapid assessment without consulting multiple departments.	High	Stages 1–6
Civil Servant	select a preset and launch simulation in one click	I can demonstrate policy trade-offs without technical setup.	High	Preset Library
Researcher	view the risk matrix for any policy	I can identify high-risk dimensions to focus my work on.	High	Stage 4
Student	run a simulation and read the final wargame report	I can study policy impact modelling interactively.	Medium	Stage 6
Any User	see a green status pill before launching	I know the system is ready before running an analysis.	High	Health Check

5.2 Use Cases

Free-Text Simulation: User types a policy description → clicks LAUNCH → backend orchestrates Stages 1–6 via GLM → results stream live to frontend → full report ready in ~60 seconds.

Preset Flow: User selects preset → policy text auto-populates → same six-stage orchestration executes. Ideal for demos and onboarding.

Error Recovery: Backend offline or key missing → status pill turns red → descriptive error shown → no partial data displayed.

6. Features Included (Scope Definition)

- Six-stage GLM-orchestrated wargame simulation with live SSE streaming.
- Free-text policy input with client-side validation (length check, token estimate).
- Five preset Malaysian policy scenarios.
- Per-stage progress cards showing status and output summary.
- Backend health check (/health) with green/red frontend status pill.
- Graceful error handling for all known failure modes.
- Hallucination mitigation: prompt anchoring, [UNVERIFIED] flagging, JSON retry logic.

7. Features Not Included (Scope Control)

- User authentication or session management.
- Persistent storage or simulation history.
- PDF/document export of reports.
- Live external data integration (DOSM, BNM, EPU APIs).
- Mobile-responsive UI or Bahasa Malaysia interface.

8. Assumptions & Constraints

LLM Cost

Estimated ~8,000–12,000 tokens per full simulation. Cost kept manageable via: 1,500-token input cap per stage, single-simulation request queue, and structured JSON prompts that minimise verbose outputs.

Technical

- Requires a valid ILMU_API_KEY in backend/.env — no key means no simulations.
- Single-threaded Node.js Express server — suitable for demo, not production-scale concurrency.
- Frontend is a static HTML file opened directly in the browser — no build step required.

Performance

- Each GLM stage takes ~4–8 seconds. Full simulation completes in 30–60 seconds.
- Concurrent requests queued; extras receive HTTP 429 until the active simulation finishes.

User Input

- English input assumed. Bahasa Malaysia accepted but GLM responds in English per system prompt.
- Input quality directly affects output quality — the system does not validate policy feasibility.

9. Risks & Questions Throughout Development

Risk / Open Question	Impact	Mitigation
GLM JSON consistency: Will GLM produce consistent structured JSON across all six stages for varied inputs?	Inconsistent JSON breaks stage-chaining and degrades downstream output quality.	Strict JSON-only system prompts + up to 2 retries + graceful degradation if retries fail.
Hallucination: Can GLM fabricate policy figures not present in the input?	Fabricated data could mislead users treating the output as factual.	System prompt anchors GLM to input-only data. Numbers not in the input are flagged [UNVERIFIED]. Report footer disclaims AI-generated nature.
Context accumulation: Will growing stage outputs exceed GLM's context window?	Later stages may lose early-stage context, producing shallow synthesis.	Token budget per stage. Stage summaries compressed before forwarding. Chunking for oversized inputs.
API availability: What if ILMU API is down during the hackathon demo?	Complete loss of simulation — the entire product depends on this external API.	Per-stage retry (max 3). Clear user-facing error states. Offline fallback plan with pre-recorded outputs.